

$$7) \vec{r} = \begin{bmatrix} 3t \\ 3t^2 \\ 2t^3 \end{bmatrix}$$

a) zapiši enačbo ploske, ki jo sest. premice, ki gredo skozi posamezne točke na krivulji v smeri $\vec{N}$.

b) poišči evoluto krivulje.

↳ krivulja, ki jo sest. središč. pritiščenih krogov.

krog, ki se dotika krivulje, vzporeden s prit. ravnino in ima polmer enak krivinskemu polmeru v dani točki
($\rho = \frac{1}{\kappa}$)

$$1) \dot{\vec{r}} = \begin{bmatrix} 3 \\ 6t \\ 6t^2 \end{bmatrix} \quad \ddot{\vec{r}} = \begin{bmatrix} 0 \\ 6 \\ 12t \end{bmatrix}$$

$$\ddot{\vec{r}} \times \dot{\vec{r}} = \begin{vmatrix} i & j & k \\ 3 & 6t & 6t^2 \\ 0 & 6 & 12t \end{vmatrix} = \begin{bmatrix} 72t^2 - 36t^2 \\ -36t \\ 18 \end{bmatrix} = 18 \begin{bmatrix} 2t^2 \\ -2t \\ 1 \end{bmatrix}$$

$$\vec{B} = \frac{\ddot{\vec{r}} \times \dot{\vec{r}}}{\|\ddot{\vec{r}} \times \dot{\vec{r}}\|} = \frac{(2t^2, -2t, 1)}{\sqrt{4t^4 + 4t^2 + 1}} = \frac{(2t^2, -2t, 1)}{2t^2 + 1}$$

$$\vec{T} = \frac{\dot{\vec{r}}}{\|\dot{\vec{r}}\|} = \frac{(1, 2t, 2t^2)}{2t^2 + 1}$$

$$1 - (2t^2)^2 = (1 + 2t^2)(1 - 2t^2)$$

$$\vec{N} = \vec{B} \times \vec{T} = \frac{1}{(2t^2 + 1)^2} \begin{vmatrix} i & j & k \\ 2t^2 & -2t & 1 \\ 1 & 2t & 2t^2 \end{vmatrix} = \frac{1}{(2t^2 + 1)^2} \begin{bmatrix} -4t^3 - 2t \\ -4t^4 + 1 \\ 4t^3 + 2t \end{bmatrix} = \frac{1}{2t^2 + 1} \begin{bmatrix} -2t \\ 1 - 2t^2 \\ 2t \end{bmatrix}$$

$$\text{normala: } \vec{R} = \vec{r}_0 + s\vec{N}(2t^2 + 1) = \begin{bmatrix} 3t \\ 3t^2 \\ 2t^3 \end{bmatrix} + s \begin{bmatrix} -2t \\ 1 - 2t^2 \\ 2t \end{bmatrix} = \begin{bmatrix} 3t - 2st \\ 3t^2 - s(1 - 2t^2) \\ 2t^3 + 2st \end{bmatrix}$$

Ko t in s pretečeta vse vrednosti, zg. enačba ↑ opiše iskano ploskev.

$$b) \rho = \frac{1}{\chi} = \frac{\|\vec{r}\|^3}{\|\vec{r} \times \ddot{\vec{r}}\|} = \frac{(3\sqrt{1+4t^2+4t^4})^3}{18\sqrt{1+4t^2+4t^4}} = \frac{3}{2}(1+2t^2)^2$$

središče prit. kroga: $\vec{R} = \vec{r} + \rho \vec{N} = \begin{bmatrix} 3t \\ 3t^2 \\ 2t^3 \end{bmatrix} + \frac{3}{2}(1+2t^2)^2 \frac{1}{1+2t^2} \begin{bmatrix} -2t \\ 1-2t^2 \\ 2t \end{bmatrix} =$

$$= \begin{bmatrix} 3t + \frac{3}{2}(1+2t^2)(-2t) \\ 3t^2 + \frac{3}{2}(1+2t^2)(1-2t^2) \\ 2t^3 + \frac{3}{2}(1+2t^2)2t \end{bmatrix} = \begin{bmatrix} -6t^3 \\ \frac{3}{2} - 6t^4 + 3t^2 \\ 3t + 8t^3 \end{bmatrix}$$

evoluta

Krivuljni integral skalarne polja

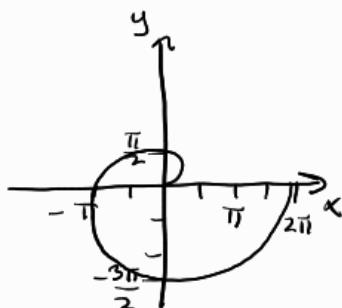
$$w = \phi(x, y, z) \quad \phi: K \rightarrow \mathbb{R}; K \subset \mathbb{R}^3 \text{ krivulja}$$

• če $\vec{r} = \vec{r}(t); t \in [a, b]$ parametrizacija, je $\int_K w ds = \int_a^b \phi(\vec{r}(t)) \cdot \|\dot{\vec{r}}(t)\| dt$

8. Izračunaj $\int_K \sqrt{x^2 + y^2 + z^2} ds$, K je prvi zavo Arhimedae spirale:

$$\begin{aligned} x &= \alpha \cos \alpha \\ y &= \alpha \sin \alpha \quad \alpha \in [0, 2\pi] \\ z &= 0 \end{aligned}$$

$$\rightarrow \vec{r} = \alpha \begin{bmatrix} \cos \alpha \\ \sin \alpha \\ 0 \end{bmatrix}$$



$$\dot{\vec{r}} = \begin{bmatrix} -\alpha \sin \alpha + \cos \alpha \\ \alpha \cos \alpha + \sin \alpha \\ 0 \end{bmatrix}$$

$$\|\dot{\vec{r}}\| = \sqrt{(\cos \alpha - \alpha \sin \alpha)^2 + (\alpha \cos \alpha + \sin \alpha)^2} =$$

$$= \sqrt{(\cos^2 \alpha - 2\alpha \sin \alpha \cos \alpha + \alpha^2 \sin^2 \alpha) + (\alpha^2 \cos^2 \alpha + 2\alpha \sin \alpha \cos \alpha + \sin^2 \alpha)}$$

$$= \sqrt{\alpha^2 + 1}$$

$$\int_K \underbrace{\sqrt{x^2 + y^2 + z^2}}_{\alpha} ds = \int_0^{2\pi} \underbrace{\alpha \sqrt{\alpha^2 + 1}}_{\sqrt{u}} d\alpha = \frac{1}{2} \int_1^{4\pi^2+1} \sqrt{u} du = \frac{1}{2} \cdot \frac{2}{3} u^{3/2} \Big|_1^{4\pi^2+1} = \frac{1}{3} ((4\pi^2+1)^{3/2} - 1)$$

$\alpha^2 + 1 = u$
 $2\alpha d\alpha = du$

Krivulja K z dolžinsko gostoto $\mu = \mu(x, y, z)$ ima

• maso $m = \int_K \mu ds$

• težišče $x^* = \frac{1}{m} \int_K \mu x ds$ $y^* = \frac{1}{m} \int_K \mu y ds$ $z^* = \frac{1}{m} \int_K \mu z ds$

• vztl. moment $J_x = \int_K (y^2 + z^2) \mu ds$ $J_y = \int_K (x^2 + z^2) \mu ds$ $J_z = \int_K (x^2 + y^2) \mu ds$

9) Izračunaj tež. in vztl. moment (J_x) krivulje $\vec{r}(t) = \begin{bmatrix} 3t \\ 3t^2 \\ 2t^3 \end{bmatrix}; t \in [0, 1];$

$\mu = \frac{y}{x+3z}$

$m = \int_K \mu ds = \int_0^1 \frac{3t^2}{3t+6t^3} \cdot 3(1+2t^2) dt =$

$\dot{\vec{r}}(t) = \begin{bmatrix} 3 \\ 6t \\ 6t^2 \end{bmatrix} \rightarrow \|\dot{\vec{r}}\| = 3(1+2t^2)$

$= 3 \int_0^1 \frac{t^2}{1+2t^2} (1+2t^2) dt = \frac{3}{2} t^2 \Big|_0^1 = \underline{\underline{\frac{3}{2}}}$

$x^* = \frac{2}{3} \int_0^1 \frac{t}{1+2t^2} \cdot 3t \cdot 3(1+2t^2) dt = 6 \cdot \frac{t^2}{2} \Big|_0^1 = \underline{\underline{2}}$

$y^* = \frac{2}{3} \int_0^1 3t \cdot 3t^2 dt = 6 \frac{t^3}{3} \Big|_0^1 = \underline{\underline{\frac{3}{2}}}$

$T^* = (2, \frac{3}{2}, \frac{4}{5})$

$z^* = \frac{2}{3} \int_0^1 3t \cdot 2t^3 dt = \frac{4}{5} t^5 \Big|_0^1 = \underline{\underline{\frac{4}{5}}}$

$J_x = \int_K \mu (y^2 + z^2) ds = \int_0^1 3t (9t^4 + 4t^6) dt = 27 \frac{t^6}{6} \Big|_0^1 + 12 \frac{t^8}{8} \Big|_0^1 = \frac{9}{2} + \frac{3}{2} = \underline{\underline{6}}$

Kriv. int. vekt. polja

$\vec{R} = \vec{\Phi}(\vec{r}); \vec{\Phi}: K \rightarrow \mathbb{R}^3; K \subset \mathbb{R}^3$ krivulja

param. kriv. $\vec{r} = \vec{r}(t), t \in [a, b]$. Definiramo

$\int_K \vec{R} \cdot d\vec{r} = \int_K \langle \vec{R}, d\vec{r} \rangle =$

$= \int_a^b \langle \vec{\Phi}(\vec{r}), \dot{\vec{r}}(t) \rangle dt$

Integral je odv. od orientacije.

10. Izračunaj $\int_{K_1} \langle \vec{R}, d\vec{r} \rangle, \int_{K_2} \langle \vec{R}, d\vec{r} \rangle$

$$R = \begin{bmatrix} y \\ -z \\ x \end{bmatrix}; K_1: \vec{r} = \begin{bmatrix} t \\ t \\ t \end{bmatrix}; K_2: \vec{r} = \begin{bmatrix} t \\ t^2 \\ t^3 \end{bmatrix}; t \in [0, 1]$$

$$\int_{K_1} \langle \vec{R}, d\vec{r} \rangle = \int_0^1 \langle (t, -t, t), (1, 1, 1) \rangle dt = \int_0^1 (t - t + t) dt = \frac{1}{2}$$

$$\int_{K_1} \langle \vec{R}, d\vec{r} \rangle = \int_0^1 \langle (t^2, -t^3, t), (1, 2t, 3t^2) \rangle dt = \int_0^1 (t^2 - 2t^4 + 3t^3) dt = \frac{1}{3} - \frac{2}{5} + \frac{3}{4} = \frac{20 - 24 + 45}{60} = \frac{41}{60}$$

Izračun po komponentah:

$$\vec{R} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} F(x, y, z) \\ G(x, y, z) \\ H(x, y, z) \end{bmatrix} \quad \vec{r} = \begin{bmatrix} f(t) \\ g(t) \\ h(t) \end{bmatrix}; t \in [a, b]$$

$$\begin{aligned} \int_K \langle \vec{R}, d\vec{r} \rangle &= \int_a^b [F(\vec{r}(t)) \cdot \dot{f}(t) + G(\vec{r}(t)) \dot{g}(t) + H(\vec{r}(t)) \dot{h}(t)] dt \\ &= \int_K \underbrace{Xdx + Ydy + Zdz}_{\text{diferencialna forma}} \end{aligned}$$

11. Izračunaj $\int_{K_1}, \int_{K_2}; K_1: \vec{r} = \begin{bmatrix} t \\ t \\ t \end{bmatrix} K_2: \vec{r} = \begin{bmatrix} t \\ t^2 \\ t^3 \end{bmatrix}; t \in [0, 1]$

$$\vec{R} = \begin{bmatrix} 2xz \\ z^2 \\ 2yz + x^2 \end{bmatrix}$$

$$\int_{K_1} \vec{R} \cdot d\vec{r} = \int_{K_1} Xdx + Ydy + Zdz = \int_0^1 2t^2 dt + t^2 dt + (2t^2 + t^2) dt = 2$$

$$\int_{K_2} \vec{R} \cdot d\vec{r} = \int_0^1 2t t^3 dt + t^6 \cdot 2t dt + (2t^2 t^3 + t^2) 3t^2 dt = 2$$